VHH analysis report

Generated 2026-10-20 15:41 · 3 sequence(s) · net charge at pH 7.4 · VHH Analyzer 20261020.1600 · TyrosinDuck

Physico-chemical properties

Name	Len	MW kDa	pI	e280	A280 1mg/mL	Instab.	Aliph.	GRAVY	Arom.	Charge	Cys
NbExample1	128	13.84	9.07	21430	1.548	40.05	64.77	-0.349	0.102	2.6	2
NbExample2	121	13.23	9.43	28420	2.148	39.96	61.32	-0.419	0.124	4.48	2
cAbLys3	133	13.99	4.68	28880	2.064	38.33	63.16	-0.228	0.12	-4.53	4

All values are theoretical, computed from the sequence alone (ProtParam). Instability index below 40 predicts a stable protein; GRAVY above -0.2 flags a relatively hydrophobic, aggregation-prone candidate.

NbExample1

Length	128 aa	MW	13.84 kDa
Theoretical pI	9.07	Net charge (pH 7.4)	+2.6
e280 (reduced)	21430 M-1cm-1	A280 at 1 mg/mL	1.548
Instability index	40.05	Aliphatic index	64.77
GRAVY	-0.349	Aromaticity	0.102

Developability flags

• Instability index ≥ 40 — predicted unstable in vitro
• 3 methionines — oxidation liability

Regions

Region	Range	Length	Sequence
FR1	1–25	25	EVQLVESGGGLVQPGGSLRLSCAAS
CDR1	26–33	8	GRTFSYNP
FR2	34–50	17	MGWFRQAPGKGRELVAA
CDR2	51–58	8	ISRTGGST
FR3	59–96	38	YYPDSVEGRFTISRDNAKRMVYLQMNSLRAEDTAVYYC
CDR3	97–117	21	AAAGVRAEDGRVRTLPSEYTF
FR4	118–128	11	WGQGTQVTVSS

Annotation method: ANARCI / IMGT

Sequence

```
EVQLVESGGGLVQPGGSLRLSCAASGRTFSYNPMGWFRQAPGKGRELVAAISRTGGSTYYPDSVEGRFTI
SRDNAKRMVYLQMNSLRAEDTAVYYCAAAGVRAEDGRVRTLPSEYTFWGQGTQVTVSS
```

VHH hallmarks — tetrad FGRL, 3/4 camelid (ANARCI / IMGT numbering)

IMGT	Kabat	Residue	VH3	Camelid	Verdict
42	37	F	V	F/H/L/Y	camelid
49	44	G	G	D/E/G/Q	VH-like
50	45	R	L	C/L/P/R	camelid
52	47	L	W	F/G/L/S/W	camelid

Indicative only: IMGT 49 is shared with VH3, and engineered or humanised constructs sit between the two profiles.

Closest germline

Species	Closest V gene	V identity	J gene	J identity
alpaca	IGHV3-3*01	84.7 %	IGHJ4*01	85.7 %
human	IGHV3-23*04	81.6 %	IGHJ5*01	71.4 %

Identity to the closest human germline is a first-pass humanness indicator, not an immunogenicity prediction.

Sequence liabilities

Liability	Motif	Position	Region	Severity
Deamidation	NS	84–85	FR3	high
Isomerisation	DS	62–63	FR3	medium
Isomerisation	DG	105–106	CDR3	medium
Deamidation (slow)	NA	74–75	FR3	low
Oxidation (Met)	M	34	FR2	low
Oxidation (Met)	M	78	FR3	low
Oxidation (Met)	M	83	FR3	low
Oxidation (Trp)	W	36	FR2	low
Oxidation (Trp)	W	118	FR4	low

A motif inside a CDR may affect binding; the same motif in a framework is mainly a stability or heterogeneity concern. Severity is a heuristic, not an experimental measurement.

NbExample2

Length	121 aa	MW	13.23 kDa
Theoretical pI	9.43	Net charge (pH 7.4)	+4.48
e280 (reduced)	28420 M-1cm-1	A280 at 1 mg/mL	2.148
Instability index	39.96	Aliphatic index	61.32
GRAVY	-0.419	Aromaticity	0.124

Developability flags

No flag raised.

Regions

Region	Range	Length	Sequence
FR1	1–25	25	QVQLQESGGGLVQAGGSLRLSCAAS
CDR1	26–33	8	GRTFSSYA
FR2	34–50	17	MGWFRQAPGKEREFVAA
CDR2	51–58	8	ISWSGGST
FR3	59–96	38	YYADSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYC
CDR3	97–110	14	AADRLSITSRRYNY
FR4	111–121	11	WGQGTQVTVSS

Annotation method: ANARCI / IMGT

Sequence

```
QVQLQESGGGLVQAGGSLRLSCAASGRTFSSYAMGWFRQAPGKEREFVAAISWSGGSTYYADSVKGRFTI
SRDNAKNTVYLQMNSLKPEDTAVYYCAADRLSITSRRYNYWGQGTQVTVSS
```

VHH hallmarks — tetrad FERF, 4/4 camelid (ANARCI / IMGT numbering)

IMGT	Kabat	Residue	VH3	Camelid	Verdict
42	37	F	V	F/H/L/Y	camelid
49	44	E	G	D/E/G/Q	camelid
50	45	R	L	C/L/P/R	camelid
52	47	F	W	F/G/L/S/W	camelid

Indicative only: IMGT 49 is shared with VH3, and engineered or humanised constructs sit between the two profiles.

Closest germline

Species	Closest V gene	V identity	J gene	J identity
alpaca	IGHV3-3*01	99.0 %	IGHJ4*01	92.9 %
human	IGHV3-23*01	83.7 %	IGHJ4*01	78.6 %

Identity to the closest human germline is a first-pass humanness indicator, not an immunogenicity prediction.

Sequence liabilities

Liability	Motif	Position	Region	Severity
Deamidation	NS	84–85	FR3	high
Isomerisation	DS	62–63	FR3	medium
Deamidation (slow)	NA	74–75	FR3	low
Deamidation (slow)	NT	77–78	FR3	low
Oxidation (Met)	M	34	FR2	low
Oxidation (Met)	M	83	FR3	low
Oxidation (Trp)	W	36	FR2	low
Oxidation (Trp)	W	53	CDR2	low
Oxidation (Trp)	W	111	FR4	low

A motif inside a CDR may affect binding; the same motif in a framework is mainly a stability or heterogeneity concern. Severity is a heuristic, not an experimental measurement.

cAbLys3

Length	133 aa	MW	13.99 kDa
Theoretical pI	4.68	Net charge (pH 7.4)	-4.53
e280 (reduced)	28880 M-1cm-1	A280 at 1 mg/mL	2.064
Instability index	38.33	Aliphatic index	63.16
GRAVY	-0.228	Aromaticity	0.12

Developability flags

• 4 cysteines — possible unpaired or extra disulfide
• 3 methionines — oxidation liability

Regions

Region	Range	Length	Sequence
FR1	1–25	25	DVQLQASGGGSVQAGGSLRLSCAAS
CDR1	26–33	8	GYTIGPYC
FR2	34–50	17	MGWFRQAPGKEREGVAA
CDR2	51–58	8	INMGGGIT
FR3	59–96	38	YYADSVKGRFTISQDNAKNTVYLLMNSLEPEDTAIYYC
CDR3	97–122	26	AADSTIYASYYECGHGLSTGGYGYDS
FR4	123–133	11	WGQGTQVTVSS

Annotation method: ANARCI / IMGT

Sequence

```
DVQLQASGGGSVQAGGSLRLSCAASGYTIGPYCMGWFRQAPGKEREGVAAINMGGGITYYADSVKGRFTI
SQDNAKNTVYLLMNSLEPEDTAIYYCAADSTIYASYYECGHGLSTGGYGYDSWGQGTQVTVSS
```

VHH hallmarks — tetrad FERG, 4/4 camelid (ANARCI / IMGT numbering)

IMGT	Kabat	Residue	VH3	Camelid	Verdict
42	37	F	V	F/H/L/Y	camelid
49	44	E	G	D/E/G/Q	camelid
50	45	R	L	C/L/P/R	camelid
52	47	G	W	F/G/L/S/W	camelid

Indicative only: IMGT 49 is shared with VH3, and engineered or humanised constructs sit between the two profiles.

Closest germline

Species	Closest V gene	V identity	J gene	J identity
alpaca	IGHV3-3*01	81.6 %	IGHJ4*01	92.9 %
human	IGHV3-23*01	71.4 %	IGHJ5*01	85.7 %

Identity to the closest human germline is a first-pass humanness indicator, not an immunogenicity prediction.

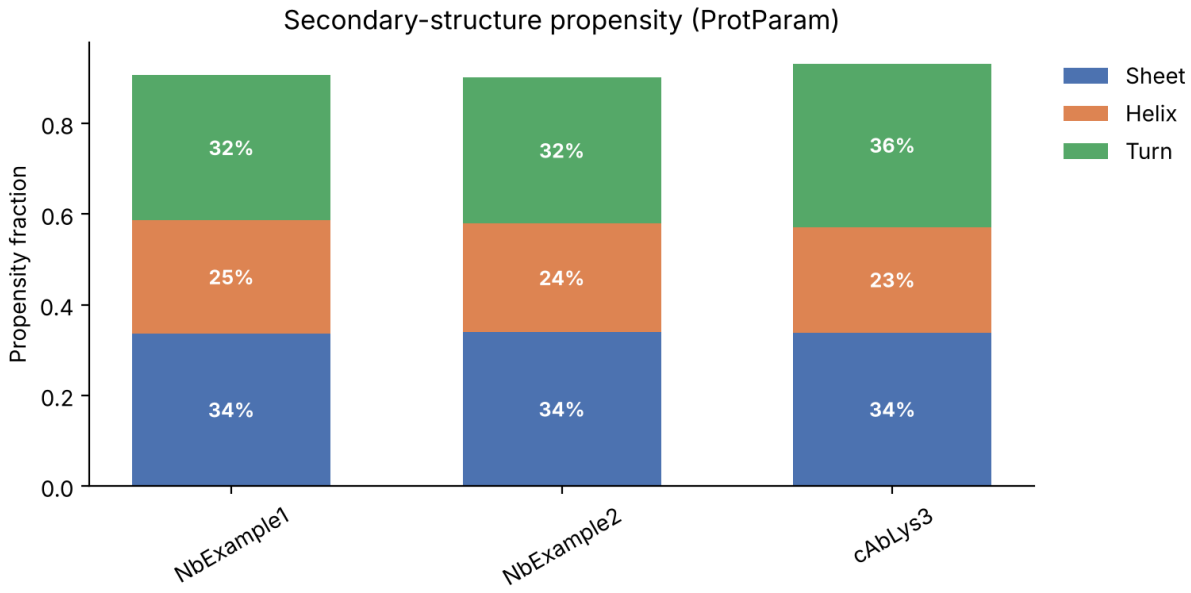
Sequence liabilities

Liability	Motif	Position	Region	Severity
Deamidation	NS	84–85	FR3	high
Extra cysteine	C	33	CDR1	high
Extra cysteine	C	109	CDR3	high
Hydrophobic patch	VYLL	79–82	FR3	medium
Isomerisation	DS	62–63	FR3	medium
Isomerisation	DS	99–100	CDR3	medium
Isomerisation	DS	121–122	CDR3	medium
Deamidation (slow)	NA	74–75	FR3	low
Deamidation (slow)	NT	77–78	FR3	low

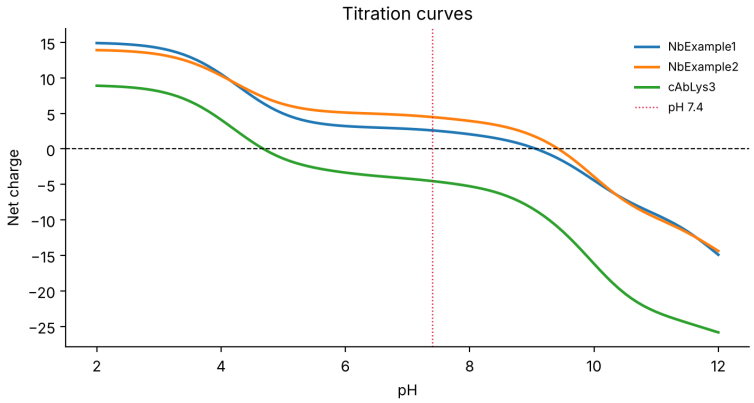
Liability	Motif	Position	Region	Severity
Oxidation (Met)	M	34	FR2	low
Oxidation (Met)	M	53	CDR2	low
Oxidation (Met)	M	83	FR3	low
Oxidation (Trp)	W	36	FR2	low
Oxidation (Trp)	W	123	FR4	low

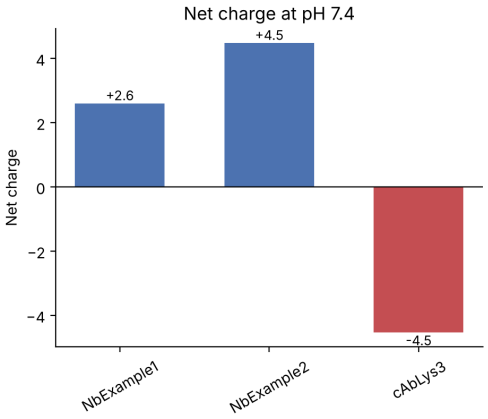
A motif inside a CDR may affect binding; the same motif in a framework is mainly a stability or heterogeneity concern. Severity is a heuristic, not an experimental measurement.

Secondary structure

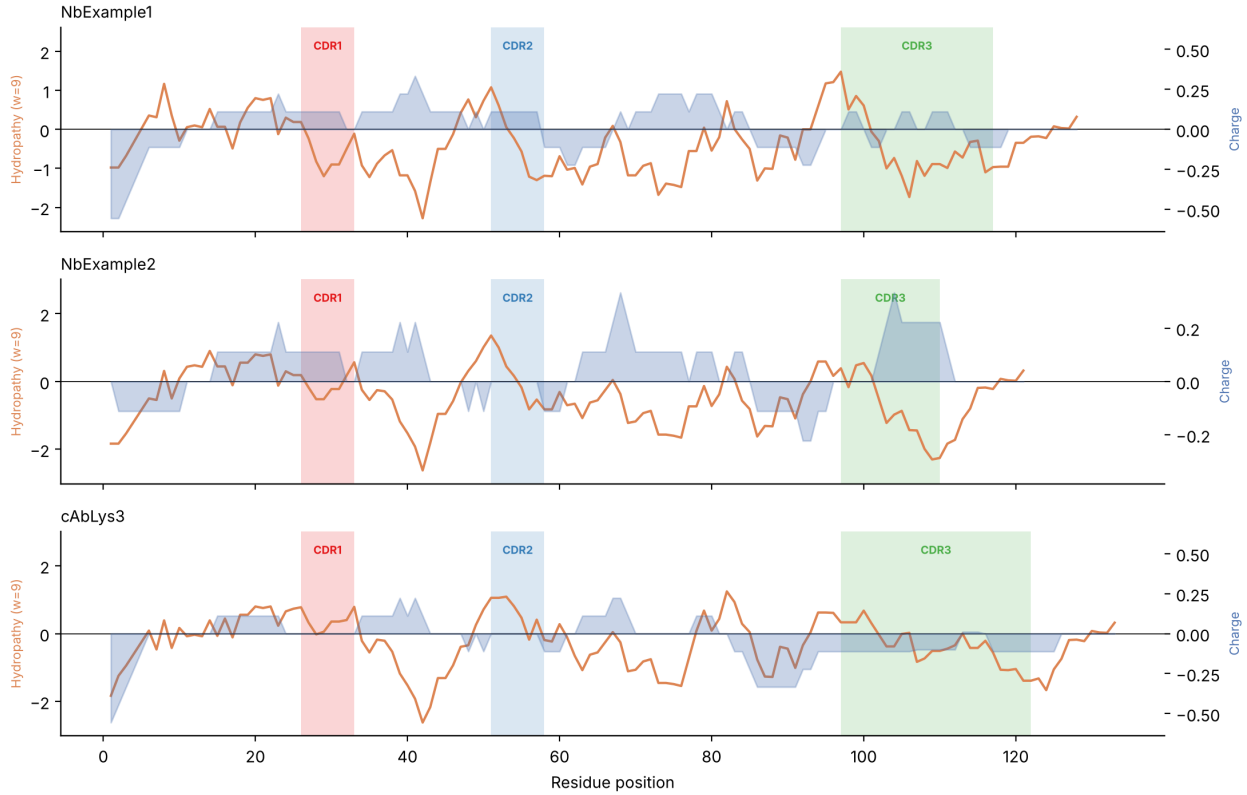


Charge

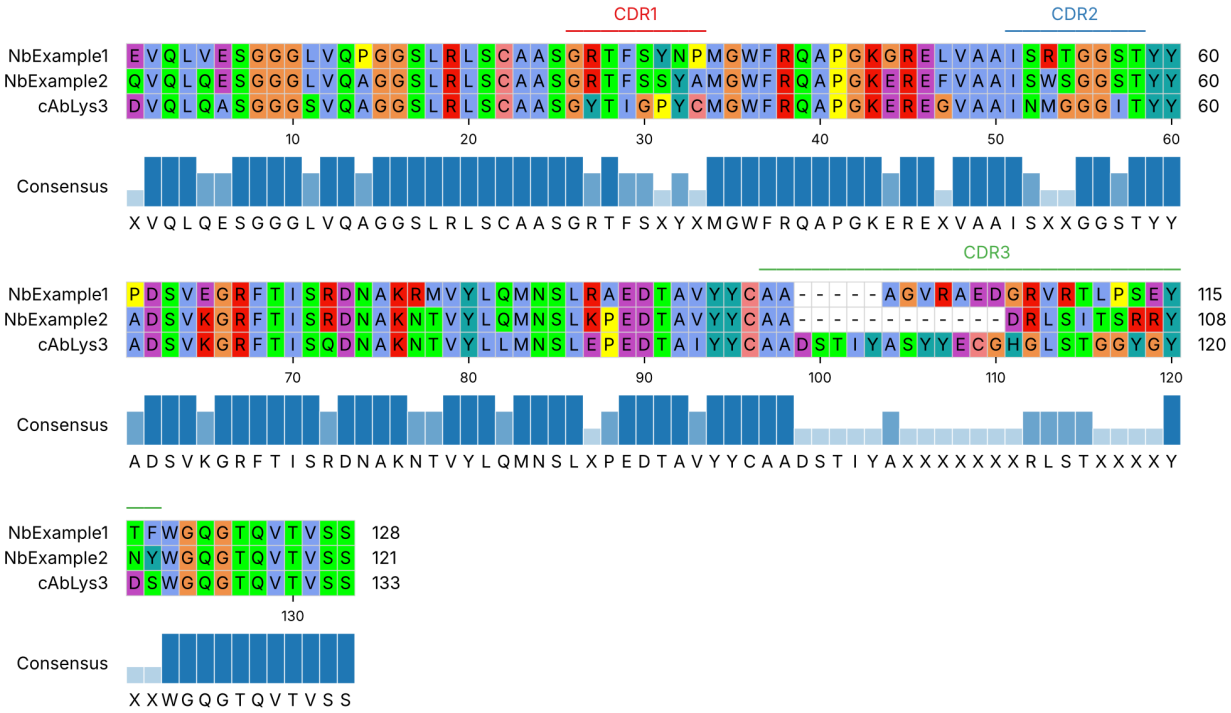

Titration curves
NbExample1
NbExample2
cAbLys3
pH 7.4
Net charge
pH
15
10
5
0
−5
−10
−15
−20
−25
2
4
6
8
10
12


Net charge at pH 7.4
Net charge
+2.6
+4.5
-4.5
4
2
0
−2
−4
NbExample1
NbExample2
cAbLys3

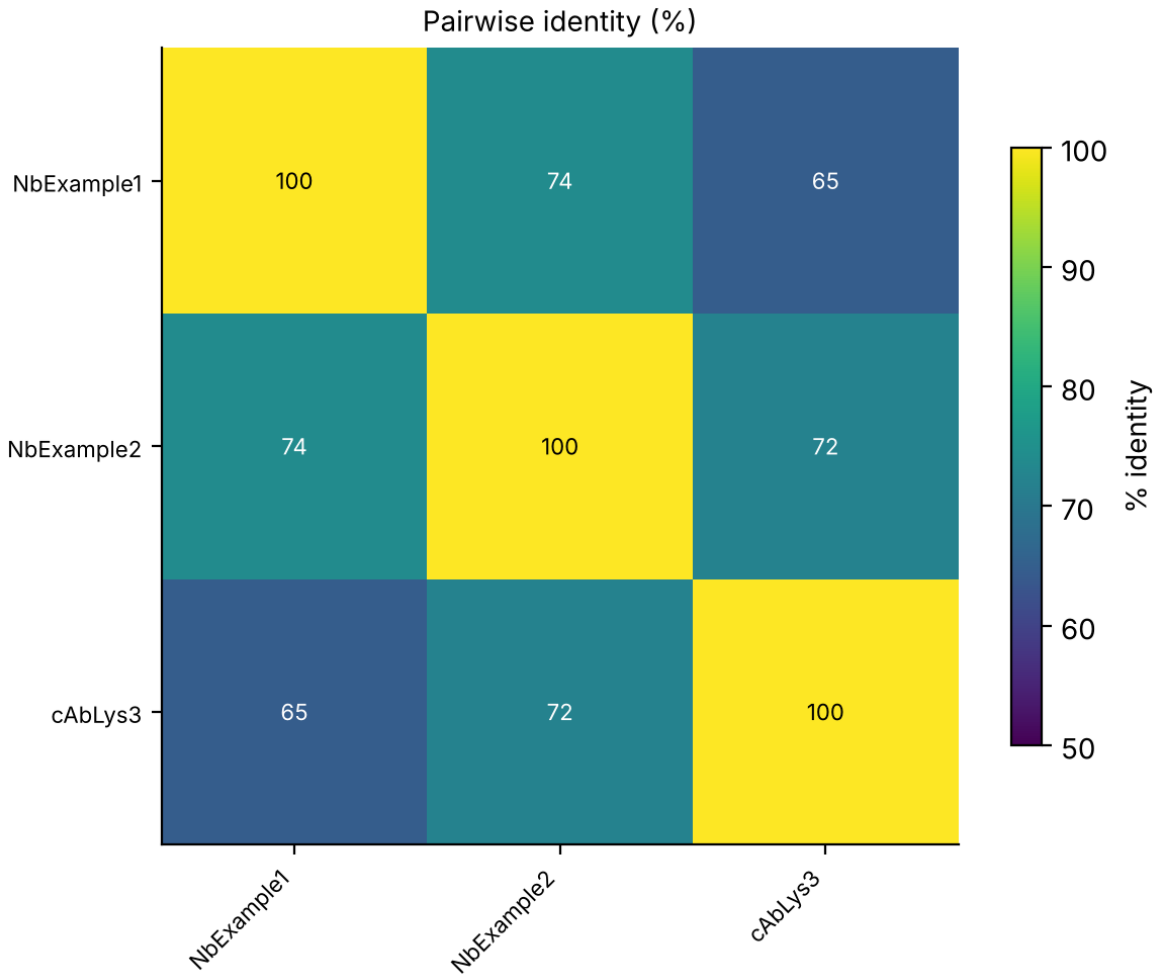
Profiles


NbExample1
CDR1
CDR2
CDR3
Hydropathy (w=9)
Charge
NbExample2
CDR1
CDR2
CDR3
Hydropathy (w=9)
Charge
cAbLys3
CDR1
CDR2
CDR3
Hydropathy (w=9)
Charge
Residue position

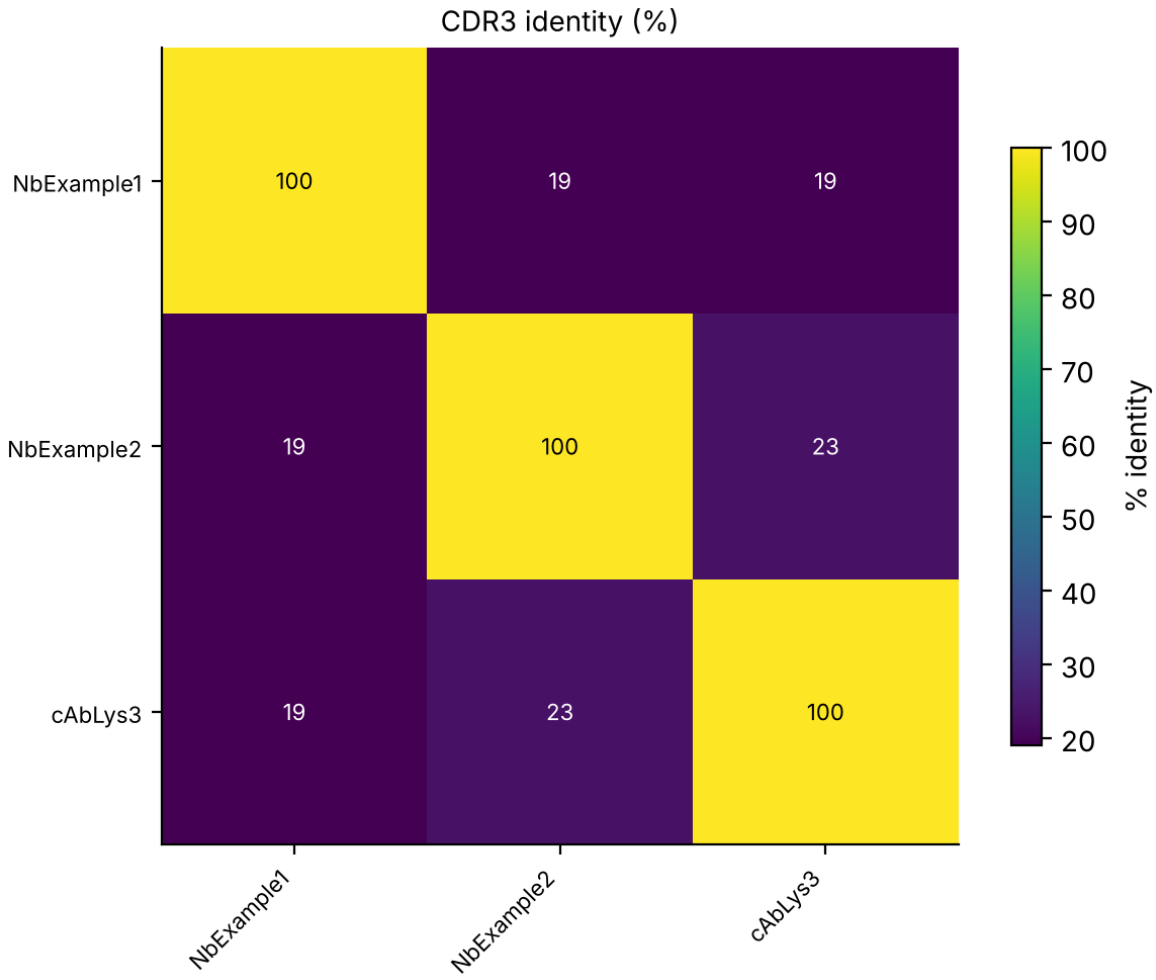
Alignment


CDR1
CDR2
NbExample1 EVQLVESGGGLVQPGGSLRLSCAASGRTFSYNPMGWFRQAPGKGRELVAAISRTGGSTYY 60
NbExample2 QVQLQESGGGLVQAGGSLRLSCAASGRTFSSYAMGWFRQAPGKEREFVAAISWSGGSTYY 60
cAbLys3 DVQLQASGGGSVQAGGSLRLSCAASGYTIGPYCMGWFRQAPGKERFGVAAINMGGGITYY 60
10 20 30 40 50 60
Consensus
XVQLQESGGGLVQAGGSLRLSCAASGRTFSXYXMGWFRQAPGKEREXVAAISXXGGSTYY
CDR3
NbExample1 PDSVEGRFTISRDNAKRMVYLQMNSLRAEDTAVYYCAA-----AGVRAEDGRVRTLPSEY 115
NbExample2 ADSVKGRFTISRDNAKNTVYLQMNSLKPEDTAVYYCAA----------DRLSITSRRY 108
cAbLys3 ADSVKGRFTISQDNAKNTVYLLMNSLEPEDTAIYYCAADSTIYASYYECGHGLSTGGYGY 120
70 80 90 100 110 120
Consensus
ADSVKGRFTISRDNAKNTVYLQMNSLXPEDTAVYYCAADSTIYAXXXXXXXXRLSTXXXXY
NbExample1 TFWGQGTQVTVSS 128
NbExample2 NYWGQGTQVTVSS 121
cAbLys3 DSWGQGTQVTVSS 133
130
Consensus
XXWGQGTQVTVSS

Identity


Pairwise identity (%)
NbExample1
NbExample2
cAbLys3
100
74
65
74
100
72
65
72
100
NbExample1
NbExample2
cAbLys3
100
90
80
70
60
50
% identity

CDR3 identity



Notes and caveats

• Every value in this report is theoretical, derived from the amino-acid sequence alone. Confirm the isoelectric point by IEF or cIEF and the extinction coefficient against a measured A280 before using them quantitatively.
• Secondary-structure fractions are composition-based propensities, not a structure prediction. Use DSSP on a model for real secondary structure.
• CDR boundaries depend on the method: IMGT windows via ANARCI are tighter than the regex anchors, so CDR1 and CDR2 lengths are not comparable between the two.
• Liability motifs and developability flags are triage heuristics. They indicate where to look, not what will happen.
• Germline identity reflects similarity to the closest germline gene; it is not an immunogenicity prediction.